

LCXCM — Liability Connection Chain Module

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Canonical Definition

The Liability Connection Chain Module (LCXCM) governs how Governed Liability Connection Relationships are assembled into ordered, traceable, bounded, temporally coherent, and reconstructable Liability Connection Chains and, where necessary, Liability Connection Networks.

LCXCM determines how individual Connection Relationships link through shared endpoints and intermediate elements, how sequence and direction are preserved, where chains begin and end, where they branch, converge, diverge, intersect, or remain incomplete, and how multiple plausible or competing chains are represented without prematurely determining their Liability significance.

LCXCM does not determine Connection Qualification, evidentiary sufficiency, Liability Attachment, or Liability.

It governs the transition:

Governed Liability Connection Relationship Set → Relationship Linking → Chain Formation → Sequence & Direction → Intermediate Element Governance → Branching & Network Structure → Chain Integrity → Governed

Liability Connection Chain Set

Its foundational distinction is:

Connection Relationship ≠ Connection Chain.

Operational Role

The operational role of LCXCM is:

to transform individually governed Liability Connection Relationships into explicit Connection Chains capable of showing how materially relevant Subjects, Objects, Conditions, conduct, events, states, consequences, and other elements are connected across the Liability inquiry.

LCXIM identifies which potential Connections require governance.

LCXRM defines the individual Relationships.

LCXCM determines how those Relationships connect.

LCXQM subsequently determines whether the resulting Connections are sufficiently qualified for Liability Determination.

Position in the Parent Standard Flow

The internal LCNS progression is:

Liability Connection Identification → Liability Connection Relationship → Liability Connection Chain → Liability Connection Qualification → Liability Connection Evidence Association LCXCM governs:

Liability Connection Chain

Its principal input is:

Governed Liability Connection Relationship Set

Its principal output is:

Governed Liability Connection Chain Set

Its canonical handoff is:

Governed Liability Connection Chain Set → Liability Connection Qualification Governance

Central Operational Question

How do the governed Liability Connection Relationships combine into complete, ordered, traceable, and materially coherent Connection Chains without collapsing distinct relationships, hiding intermediate elements, or assuming Liability?

Governed Space

LCXCM governs:

Connection Chain Identification, Connection Chain Identity, Relationship Linking, Chain Origin, Chain Endpoint, Chain Direction, Chain Sequence, Intermediate Elements, Intermediate Relationships, Direct Chains, Indirect Chains, Multi-Step Chains, Parallel Chains, Converging Chains, Diverging Chains, Branching Chains, Merged Chains, Nested Chains, Dependent Chains, Alternative Chains, Competing Chains, Partial Chains, Broken Chains, Circular Structures, Connection Networks, Chain Boundaries, Chain Temporal Structure, Chain Integrity, Chain Continuity, Chain Change, Chain Correction, Chain Reassessment, and Chain Sufficiency for Qualification.

Liability Connection Chain

A Liability Connection Chain is a governed sequence of Connection Relationships linking materially relevant elements within a Liability inquiry.

Canonical form:

$A \rightarrow B \rightarrow C$ or:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ Each arrow represents a governed Connection Relationship.

Each node represents a governed element.

The chain represents their connective structure.

It does not represent Liability itself.

Therefore:

Liability Connection Chain $\neq$ Liability.

Chain Formation

Chain Formation occurs when individually governed Connection Relationships can be linked through common or legitimately related endpoints.

Example:

$A \rightarrow B$ and:

$B \rightarrow C$ may form:

$A \rightarrow B \rightarrow C$ only where the identity and relational continuity of B are sufficiently established.

Shared Endpoint Principle

Relationships should not be joined merely because their endpoint labels appear similar.

The shared element must be governably continuous or otherwise legitimately related.

Therefore:

Same Endpoint Label $\neq$ Same Governed Endpoint.

Chain Identity

Each materially distinct Connection Chain should possess a governed identity where necessary for: reference, qualification, evidence association, comparison, change tracking, and historical reconstruction.

Therefore:

Chain Identity $\neq$ Relationship Identity.

Chain Identity and Liability

A Chain Identity does not represent a Liability identity.

Therefore:

Connection Chain Identity $\neq$ Liability Identity.

Chain Origin

The Chain Origin identifies the materially relevant starting element of a Connection Chain.

The origin depends upon the governance question.

It must not be assumed to be:

the first chronologically observed event, the first recorded event, or the first element discovered.

Therefore:

First Observed Element $\neq$ Chain Origin by default.

Chain Endpoint

The Chain Endpoint identifies the materially relevant terminal element of the governed chain for the specific Liability inquiry.

An endpoint may be:

Object state, Condition-relevant state, event, Harm state, consequence, or another governed element.

Therefore:

Chain Endpoint $\neq$ Liability Conclusion.

Chain Direction

The direction of individual Relationships should be preserved throughout the Chain.

Canonical representation:

$A \rightarrow B \rightarrow C$ should not be converted into:

$C \rightarrow B \rightarrow A$ without a governed basis.

Therefore:

Chain Direction $\neq$ Presentation Order.

Bidirectional Components

A chain may contain a bidirectional Relationship.

Example:

$A \leftrightarrow B \rightarrow C$ The bidirectional component should remain explicit.

It must not make the entire chain bidirectional automatically.

Unknown Direction Component

Where the direction of one Relationship remains unknown, the Chain should preserve that uncertainty.

Example:

$A \rightarrow B ? C \rightarrow D$ The system should not manufacture direction merely to complete the Chain.

Chain Sequence

LCXCM preserves the materially relevant sequence of Relationships.

Where sequence affects meaning:

$A \rightarrow B \rightarrow C$ is distinct from:

$A \rightarrow C \rightarrow B$ Therefore:

Same Elements $\neq$ Same Connection Chain.

Temporal Sequence

Chain sequence may depend on time.

LCXCM should preserve, where relevant:

event order, state transitions, activation periods, delays, overlaps, and temporal uncertainty.

Logical Sequence

Not every Connection Chain is purely chronological.

A Chain may contain logical or structural dependencies.

Therefore:

Connection Sequence $\neq$ Chronological Sequence by default.

Both may need separate representation.

Intermediate Element

An Intermediate Element is any materially relevant governed element situated between the Chain Origin and Endpoint.

Intermediate elements may include:

Subjects, Objects, Conditions, actions, omissions, events, states, decisions, systems, mechanisms, Harm states, or consequences.

Intermediate Element Preservation

A materially relevant Intermediate Element should not be removed merely to simplify the Chain.

Therefore:

Simplified Chain $\neq$ Governed Complete Chain by default.

Intermediate Relationship

Each pair of linked elements should possess an identifiable governed Relationship where that relationship is materially necessary.

A chain must not rely on invisible relational jumps.

Therefore:

Adjacent Nodes $\neq$ Established Relationship by default.

Hidden Intermediate Element

A later analysis may reveal an Intermediate Element omitted from the original Chain.

The Chain should be capable of revision without erasing its earlier representation.

Therefore:

Discovered Intermediate Element $\neq$ New Liability Origin.

Direct Chain

A Direct Chain may contain a single governed Relationship:

$A \rightarrow B$ Directness refers to the absence of a materially necessary intermediate element for the specific inquiry.

Therefore:

Direct Chain $\neq$ Strong Connection by default.

Indirect Chain

An Indirect Chain contains one or more materially necessary Intermediate Elements.

Example:

$A \rightarrow B \rightarrow C$ Indirectness does not make the Chain invalid.

Therefore:

Indirect Chain $\neq$ Weak Chain by default.

Multi-Step Chain

A Multi-Step Chain contains several sequential Relationships.

Example:

Subject $\rightarrow$ Decision $\rightarrow$ Action $\rightarrow$ Event $\rightarrow$ Object State $\rightarrow$ Consequence Each materially relevant step should remain separately governable.

Chain Compression

A Multi-Step Chain should not be compressed into:

Subject $\rightarrow$ Consequence where the omitted steps materially affect governance.

Therefore:

Chain Compression $\neq$ Chain Simplification where material structure is lost.

Parallel Chains

Two or more Connection Chains may operate independently within the same Liability matter.

Example:

Subject A $\rightarrow$ Action A $\rightarrow$ Event Subject B $\rightarrow$ Action B $\rightarrow$ Event Both should remain independently identifiable.

Parallelism and Liability

Parallel Chains do not imply equal Liability relevance.

Therefore:

Parallel Connection Chains $\neq$ Equal Liability Significance.

Qualification and Determination remain downstream.

Converging Chains

Multiple Chains may converge on one element.

Example:

$A \rightarrow X$ $B \rightarrow X$ $C \rightarrow X$ Convergence should preserve the independent upstream paths.

Therefore:

Convergence $\neq$ Common Liability by default.

Diverging Chains

One Chain may divide into several downstream paths.

Example:

$A \rightarrow B \rightarrow C$ and then:

$C \rightarrow D$ $C \rightarrow E$ $C \rightarrow F$ Each branch should remain independently traceable.

Branching Chain

A Branching Chain contains a point from which multiple materially relevant Connection paths emerge.

The branch point should retain identity and temporal position.

Therefore:

One Origin $\neq$ One Endpoint.

Merged Chains

Previously separate Chains may merge where they reach a common governed element.

The upstream Chains should not lose their identities merely because they merge.

Therefore:

Chain Merger $\neq$ Historical Chain Erasure.

Nested Chain

A complex Connection Chain may contain a subordinate Chain that itself requires governance.

Nested structures may be used where they improve clarity without hiding material relationships.

Dependent Chain

One Connection Chain may depend upon another Chain being established.

Therefore:

Dependent Chain $\neq$ Independent Chain.

Dependency should remain explicit.

Alternative Chain

Where available information supports more than one plausible sequence, LCXCM should preserve alternative Chains.

Example:

$A \rightarrow B \rightarrow D$ or:

$A \rightarrow C \rightarrow D$ Therefore:

Alternative Chain $\neq$ Established Preferred Chain.

Competing Chains

Competing Chains exist where different governed representations cannot simultaneously describe the same relevant sequence in the same manner.

LCXCM should preserve the conflict.

Therefore:

Competing Chains $\neq$ Authority to Select the Convenient Chain.

Partial Chain

A Partial Chain exists where only part of the relevant connective sequence can be governed.

Example:

$A \rightarrow B \rightarrow ? \rightarrow D$ Therefore:

Partial Chain $\neq$ Complete Chain.

Broken Chain

A Broken Chain contains a material gap between Relationships that cannot presently be governed as connected.

Example:

$A \rightarrow B$ [gap] $C \rightarrow D$ A Broken Chain should not be silently completed through assumption.

Therefore:

Broken Chain $\neq$ Complete Connection Chain.

Unknown Chain Segment

A Chain may contain an unknown segment where sufficient reason exists to believe an intermediate structure may exist but it cannot yet be governed.

Unknown segments should remain explicit.

Therefore:

Unknown Segment $\neq$ Established Intermediate Relationship.

Inferred Chain Segment

A Chain segment may be inferred where the methodology permits inference.

The inference should preserve:

supporting inputs, logic, assumptions, and uncertainty.

Therefore:

Inferred Chain Segment $\neq$ Directly Established Chain Segment.

Assumed Chain Segment

An assumed segment may be used for explicitly hypothetical analysis.

It must not be represented as established.

Therefore:

Assumed Chain Segment $\neq$ Governed Established Chain Segment.

Circular Structure

Some relational structures may appear circular:

$A \rightarrow B \rightarrow C \rightarrow A$ LCXCM should preserve genuine circularity where it exists.

It should not create circularity merely because the same elements recur.

Circularity and Duplication

Repeated appearance of an element does not automatically establish a circular Chain.

Therefore:

Repeated Element $\neq$ Circular Connection Chain.

Connection Network

Where the Liability matter contains multiple intersecting Chains, LCXCM may represent them as a:

Liability Connection Network

A Network may contain:

nodes, Relationships, Chains, branches, convergences, divergences, dependencies, and alternative paths.

Chain and Network

A Chain is a governed path through the Connection structure.

A Network is the broader structure within which multiple Chains may exist.

Therefore:

Connection Chain $\neq$ Connection Network.

Network Node

A Network Node represents a governed element participating in one or more Connection Relationships or Chains.

One node may participate in several distinct paths.

Therefore:

Shared Node $\neq$ Shared Chain Identity.

Network Edge

A Network Edge represents a governed Connection Relationship.

The Network should not create an edge merely because two nodes coexist.

Therefore:

Node Coexistence $\neq$ Network Edge.

Network Path

A Network may contain multiple paths between the same origin and endpoint.

Those paths may possess different:

Relationships, intermediate elements, timing, uncertainty, and later qualification.

Therefore:

Same Origin and Endpoint $\neq$ Same Connection Path.

Multi-Subject Chain

A Connection Chain may contain several Subjects.

Example:

Subject A $\rightarrow$ Subject B $\rightarrow$ Action $\rightarrow$ Event The presence of multiple Subjects does not establish shared Liability.

Multi-Object Chain

A Chain may involve several Liability Objects or Object components.

Object transitions should remain explicit where they materially affect the Connection structure.

Condition-Integrated Chain

A Liability Condition may form part of a Chain where its relationship to a Subject, action, state, or event is materially relevant.

Example:

Subject $\rightarrow$ Conduct $\rightarrow$ Condition-Relevant State $\rightarrow$ Event The Condition's State remains governed by LCOS.

Basis-Specific Chain

A Chain may be relevant under one Liability Basis and not another.

Therefore:

Connection Chain under Basis A $\neq$ Connection Chain under Basis B by default.

Basis association should remain explicit.

Relationship-Specific Chain

Different upstream Liability Relationships may produce different Connection Chains even for the same Subject and Object.

Therefore:

Same Subject + Same Object $\neq$ Same Connection Chain.

Harm-Integrated Chain

Where Harm is relevant:

Action $\rightarrow$ Event $\rightarrow$ Harm State may form part of a Liability Connection Chain.

HAGA™ governs the Harm state.

LCXCM governs only its position within the Liability Connection structure.

Therefore:

Harm Chain Element $\neq$ Liability Determination.

Operational Mechanism in a Chain

An operational mechanism may appear as an Intermediate Element.

Its presence does not transfer governance ownership of the mechanism to LIGA™.

LCXCM governs only its position and relevance within the Liability Connection Chain.

Physical Contact in a Chain

Physical contact may appear as an operational mechanism:

Action $\rightarrow$ Physical Contact $\rightarrow$ Operational State or within another relevant sequence.

Physical contact remains an operational mechanism.

Therefore:

Physical Contact $\neq$ Harm $\neq$ Liability.

Operational Causation Chain

An operational architecture may establish a causal sequence.

That sequence may become an input to LCXCM.

However:

Operational Causation Chain $\neq$ Liability Connection Chain by default.

LCXCM governs the Liability-specific connective structure.

Factual Chain

A factual sequence may establish that events occurred in a particular order.

That factual sequence does not automatically become a Liability Connection Chain.

Therefore:

Factual Sequence $\neq$ Liability Connection Chain by default.

Evidence Chain

Evidence may itself possess provenance or evidentiary chains.

Those are not Liability Connection Chains.

Therefore:

Evidence Chain $\neq$ Liability Connection Chain.

ADIT® governs evidence structures.

Accountability Chain

AGA™ may govern chains of Responsibility, Authority, Accountability, or Attribution.

Those chains may become relevant inputs.

Therefore:

Accountability Chain ≠ Liability Connection Chain.

Identity Chain

OBIDENTITY® may govern Identity, Ownership, Authority, Permission, Provenance, Transfer, and Continuity chains.

Those structures may support endpoint or continuity governance.

Therefore:

Identity Chain ≠ Liability Connection Chain.

Temporal Chain Structure

LCXCM should preserve materially relevant time relationships among Chain elements.

This may include:

absolute time, relative order, duration, delay, overlap, activation, expiration, or temporal uncertainty.

Time Gap

A time gap between Chain elements does not automatically break the Chain.

Therefore:

Temporal Gap ≠ Chain Break by default.

The material relevance of the gap must be governed.

Temporal Proximity

Events occurring close together do not automatically form a Chain.

Therefore:

Temporal Proximity ≠ Connection Chain.

Spatial Proximity

Elements occurring in the same location do not automatically form a Chain.

Therefore:

Spatial Proximity ≠ Connection Chain.

Organizational Proximity

Subjects within the same organization do not automatically belong to the same Connection Chain.

Therefore:

Organizational Proximity ≠ Connection Chain.

Technical Proximity

Components operating within the same technical system do not automatically form a Liability Connection Chain.

Therefore:

Technical Proximity ≠ Connection Chain.

Chain Boundary

Every Connection Chain should possess a boundary appropriate to the Liability inquiry.

The boundary determines:

where the Chain begins, where it ends, which intermediate elements belong, and which adjacent structures remain outside the Chain.

Over-Broad Chain

A Chain may become over-broad where unrelated or immaterial elements are included.

Over-broad Chains can obscure the relevant Connection structure.

Therefore:

More Elements $\neq$ Better Chain.

Under-Specified Chain

A Chain may be under-specified where materially necessary intermediate elements are omitted.

Therefore:

Shorter Chain $\neq$ More Accurate Chain.

The appropriate chain contains the shortest structure that preserves all materially relevant Connections.

Chain Integrity

A Connection Chain possesses governance integrity when its materially relevant:

nodes, Relationships, direction, sequence, boundaries, temporal structure, and uncertainties are explicitly preserved.

Chain Integrity and Truth

Chain Integrity does not mean every Relationship has been proven conclusively.

It means the Chain representation faithfully distinguishes:

established, unknown, uncertain, disputed, inferred, and assumed components.

Therefore:

Chain Integrity $\neq$ Absolute Certainty.

Chain Completeness

A Chain is complete for its defined governance boundary when no materially necessary intermediate Relationship or element remains omitted within that boundary.

Therefore:

Chain Completeness $\neq$ Liability Sufficiency.

Chain Sufficiency

Chain Sufficiency asks whether the Chain contains enough governed structure to proceed to Connection Qualification.

It does not determine whether the Connection will qualify.

Therefore:

Chain Sufficiency $\neq$ Connection Qualification.

Chain Change

A Connection Chain may change because:

a Relationship changes, an Intermediate Element is discovered, a sequence is corrected, an endpoint changes, or new governed information becomes available.

Changes should remain traceable.

Chain Correction

A Chain representation may later be found incorrect.

Correction should preserve:

the earlier Chain, the corrected Chain, the reason, the affected Relationships, and relevant provenance.

Therefore:

Chain Correction ≠ Historical Erasure.

Chain Reassessment

New information may require reassessment of:

Chain boundaries, sequence, direction, intermediate elements, branching, or completeness.

Therefore:

Chain Reassessment ≠ Historical Rewrite.

Chain Versioning

Material changes may require a new Chain version.

Therefore:

New Chain Version ≠ New Chain Identity by default.

Continuity should be preserved where the underlying governed Chain remains the same.

Chain Split

A Chain may later be determined to contain two or more distinct Chains.

The resulting structures should preserve their common historical origin where relevant.

Therefore:

Chain Split ≠ Independent Historical Origin by default.

Chain Merger

Two Chains may later be determined to form one integrated Chain.

Their earlier identities should remain reconstructable.

Therefore:

Chain Merger ≠ Historical Identity Erasure.

Chain Re-Entry

Later LCNS or LIGA™ stages may reveal a missing or incorrect Chain component.

LCXCM should support targeted re-entry.

Therefore:

Chain Re-Entry ≠ New Liability Origin by default.

Upstream Re-Entry

Chain governance may reveal that:

a Connection Candidate is missing → return to LCXIM; a Relationship is incorrectly structured → return to LCXRM; an upstream Condition is incorrect → return to LCOS; or another earlier LIGA™ structure requires reassessment.

LCXCM should not repair upstream governance by inventing Chain elements.

No Outcome-to-Chain Shortcut

An outcome does not establish the Chain that produced or relates to it.

Therefore:

Outcome ≠ Connection Chain.

No Harm-to-Chain Shortcut

Established Harm does not identify the complete Liability Connection Chain.

Therefore:

Harm ≠ Liability Connection Chain.

No Causation-to-Liability Shortcut

Even a complete operational causal chain does not establish Liability.

Therefore:

Operational Causal Chain ≠ Liability Determination.

No Relationship-to-Chain Shortcut

One Connection Relationship does not establish a multi-step Chain.

Therefore:

One Relationship ≠ Complete Connection Chain by default.

No Missing-Link Assumption

A missing Chain segment must not be filled merely because doing so produces a coherent narrative.

Therefore:

Narrative Coherence ≠ Governed Chain Continuity.

No Single-Chain Assumption

A Liability matter should not be forced into one Chain where multiple materially plausible Chains exist.

Therefore:

One Matter ≠ One Connection Chain.

No Shortest-Chain Bias

The shortest available path should not automatically be treated as the relevant Liability Connection Chain.

Therefore:

Shortest Path ≠ Governing Connection Path by default.

No Longest-Chain Bias

Likewise, adding more elements does not improve the Chain automatically.

Therefore:

Longest Path ≠ Most Complete Governed Chain by default.

No Dominant-Actor Assumption

The most visible or operationally dominant Subject should not automatically become the Chain Origin.

Therefore:

Dominant Subject ≠ Chain Origin by default.

No Final-Outcome Bias

Chain construction should not work backward from a desired Liability outcome and select only Relationships supporting that outcome.

Therefore:

Desired Determination $\neq$ Valid Chain Construction Principle.

Chain Provenance

LCXCM should preserve sufficient provenance to determine:

which Relationships formed the Chain, where those Relationships originated, which transformations occurred, which components were added or removed, and how the current Chain representation developed.

Historical Chain Reconstruction

The architecture should support reconstruction of the Connection Chain as it:

existed, was represented, or was known at materially relevant lifecycle moments.

These states may differ.

Therefore:

Historical Chain Reality $\neq$ Historical Chain Knowledge by default.

Later Knowledge

A Chain may be reconstructed differently after new information appears.

LCXCM should distinguish:

what was known at T1, what was discovered at T2, and what the reassessed Chain represents.

Therefore:

Later Chain Knowledge $\neq$ Earlier Chain Knowledge.

Chain Readiness for Qualification

A Connection Chain is ready for LCXQM when it is sufficiently:

identified, bounded, ordered, directionally represented, temporally coherent, linked through governed Relationships, transparent about missing elements, and reconstructable.

Readiness does not mean Qualification has succeeded.

Governed Liability Connection Chain Set

The principal output of LCXCM is:

Governed Liability Connection Chain Set

It may contain:

Direct Chains, Indirect Chains, Multi-Step Chains, Parallel Chains, Converging Chains, Diverging Chains, Branching Chains, Merged Chains, Nested Chains, Dependent Chains, Alternative Chains, Competing Chains, Partial Chains, Broken Chains, and Connection Network structures.

Handoff to Liability Connection Qualification

LCXCM hands the:

Governed Liability Connection Chain Set

to:

LCXQM — Liability Connection Qualification Module

The next governance question becomes:

Which governed Liability Connection Chains are sufficiently relevant, coherent, supported, and methodologically qualified to proceed toward Liability Determination?

LCXCM does not answer that question.

Minimum Implementation Framework

A conforming implementation of LCXCM must provide exactly the following eight capabilities:

Chain Formation & Identity

Capability to assemble governed Connection Relationships into explicitly identified Connection Chains while preserving Relationship and Chain identities.

Origin, Endpoint, Direction & Sequence Governance

Capability to govern Chain origins, endpoints, direction, ordering, chronological and logical sequence, and unknown directional states.

Intermediate Element & Relationship Governance

Capability to preserve materially necessary Intermediate Elements and Relationships without hiding gaps or compressing meaningful structure.

Multi-Step, Parallel, Branching, Converging & Network Governance

Capability to represent complex Connection structures containing multiple paths, branches, convergences, divergences, dependencies, and networks.

Alternative, Competing, Partial & Broken Chain Governance

Capability to preserve alternative, competing, incomplete, broken, unknown, inferred, and assumed Chain structures without manufacturing completeness or certainty.

Temporal, Boundary & Integrity Governance

Capability to preserve materially relevant timing, Chain boundaries, structural integrity, completeness states, and explicit uncertainty.

Chain Change, Correction, Versioning & Re-Entry

Capability to govern Chain changes, corrections, reassessments, versions, splits, mergers, historical reconstruction, and targeted lifecycle re-entry without historical erasure.

Chain Sufficiency & Handoff

Capability to determine whether the Governed Liability Connection Chain Set is sufficiently structured for Connection Qualification without prematurely determining Qualification, evidentiary sufficiency, Liability Attachment, or Liability.

Governance Logic

The canonical LCXCM governance logic is:

Governed Liability Connection Relationship Set

↓

Identify Linkable Relationships

↓

Verify Shared Endpoints

↓

Establish Chain Identity

↓

Establish Chain Origin

↓

Establish Chain Endpoint

↓

Preserve Relationship Direction

↓

Determine Material Sequence

↓

Identify Intermediate Elements

↓

Identify Intermediate Relationships

↓

Construct Direct / Indirect / Multi-Step Chains

↓

Identify Parallel / Converging / Diverging / Branching Structures

↓ Construct Connection Network where required ↓

Preserve Alternative / Competing Chains

↓

Preserve Partial / Broken / Unknown Segments

↓

Establish Temporal Structure

↓

Establish Chain Boundary

↓

Assess Chain Integrity

↓

Preserve Change / Correction / Reassessment / Version History

↓

Assess Chain Sufficiency

↓

Governed Liability Connection Chain Set

↓

Handoff to Liability Connection Qualification Governance

Governance Outputs

LCXCM may produce:

Liability Connection Chain Identity

Chain Origin
Chain Endpoint
Chain Direction
Chain Sequence
Intermediate Element Set
Intermediate Relationship Set
Direct Liability Connection Chain
Indirect Liability Connection Chain
Multi-Step Liability Connection Chain
Parallel Connection Chain Set
Converging Connection Structure
Diverging Connection Structure
Branching Connection Chain
Merged Connection Chain
Nested Connection Chain
Dependent Connection Chain
Alternative Connection Chain
Competing Connection Chain
Partial Connection Chain
Broken Connection Chain
Unknown Chain Segment
Inferred Chain Segment
Assumed Chain Segment
Circular Connection Structure
Liability Connection Network
Connection Network Node
Connection Network Edge
Connection Network Path
Temporal Chain Structure
Chain Boundary

Chain Integrity State

Chain Completeness State

Chain Correction Record

Chain Reassessment Record

Chain Version Record

Historical Chain Record

Chain Sufficiency State

The principal output is:

Governed Liability Connection Chain Set

Critical Distinctions

LCXCM preserves the following distinctions:

Connection Relationship ≠ Connection Chain Liability Connection Chain ≠ Liability Same Endpoint Label ≠ Same Governed Endpoint Chain Identity ≠ Relationship Identity Connection Chain Identity ≠ Liability Identity First Observed Element ≠ Chain Origin by default Chain Endpoint ≠ Liability Conclusion Chain Direction ≠ Presentation Order Same Elements ≠ Same Connection Chain Connection Sequence ≠ Chronological Sequence by default Simplified Chain ≠ Governed Complete Chain by default Adjacent Nodes ≠ Established Relationship by default Discovered Intermediate Element ≠ New Liability Origin Direct Chain ≠ Strong Connection by default Indirect Chain ≠ Weak Chain by default Parallel Connection Chains ≠ Equal Liability Significance Convergence ≠ Common Liability by default One Origin ≠ One Endpoint Chain Merger ≠ Historical Chain Erasure Dependent Chain ≠ Independent Chain Alternative Chain ≠ Established Preferred Chain Competing Chains ≠ Authority to Select the Convenient Chain Partial Chain ≠ Complete Chain Broken Chain ≠ Complete Connection Chain Unknown Segment ≠ Established Intermediate Relationship Inferred Chain Segment ≠ Directly Established Chain Segment Assumed Chain Segment ≠ Governed Established Chain Segment Repeated Element ≠ Circular Connection Chain Connection Chain ≠ Connection Network Shared Node ≠ Shared Chain Identity Node Coexistence ≠ Network Edge Same Origin and Endpoint ≠ Same Connection Path Connection Chain under Basis A ≠ Connection Chain under Basis B by default Same Subject + Same Object ≠ Same Connection Chain Harm Chain Element ≠ Liability Determination Physical Contact ≠ Harm ≠ Liability Operational Causation Chain ≠ Liability Connection Chain by default Factual Sequence ≠ Liability Connection Chain by default Evidence Chain ≠ Liability Connection Chain Accountability Chain ≠ Liability Connection Chain Identity Chain ≠ Liability Connection Chain Temporal Gap ≠ Chain Break by default Temporal Proximity ≠ Connection Chain Spatial Proximity ≠ Connection Chain Organizational Proximity ≠ Connection Chain Technical Proximity ≠ Connection Chain More Elements ≠ Better Chain Shorter Chain ≠ More Accurate Chain Chain Integrity ≠ Absolute Certainty Chain Completeness ≠ Liability Sufficiency Chain Sufficiency ≠ Connection Qualification Chain Correction ≠ Historical Erasure Chain Reassessment ≠ Historical Rewrite New Chain Version ≠ New Chain Identity by default Chain Split ≠ Independent Historical Origin by default Chain Merger ≠ Historical Identity Erasure Chain Re-Entry ≠ New Liability Origin by default Outcome ≠ Connection Chain Harm ≠ Liability Connection Chain Operational Causal Chain ≠ Liability Determination One Relationship ≠ Complete Connection Chain by default Narrative Coherence ≠ Governed Chain Continuity One Matter ≠ One Connection Chain Shortest Path ≠ Governing Connection Path by default Longest Path ≠ Most Complete Governed Chain by default Dominant Subject ≠ Chain Origin by default Desired Determination ≠ Valid Chain Construction Principle Historical Chain Reality ≠ Historical Chain Knowledge by default Later Chain Knowledge ≠ Earlier Chain Knowledge

Cross-Architecture Boundary

LCXCM may incorporate governed elements and relationships originating in other OOF® architectures.

ORA™ may provide Operational Reality, event sequences, and operational mechanisms.

HAGA™ may provide governed Harm states and Harm lifecycle transitions.

AGA™ may provide Responsibility, Accountability, Authority, and Attribution structures.

GOA™ may provide Authority, Scope, and Delegation states.

ADIT® may provide evidence and provenance structures.

VALIDOS™ may provide validated inputs.

OBIDENTITY® may provide Identity, Ownership, Authority, Permission, Provenance, Transfer, and Continuity structures.

AIG® and ASGA™ may provide AI or autonomous-system actions, states, authority, execution, permission, and override structures.

LCXCM does not absorb those architectures' governance domains.

It governs only how materially relevant externally governed elements participate within the Liability Connection Chain.

The canonical interoperability principle is:

An externally governed mechanism, state, relationship, or lifecycle may form an element within a Liability Connection Chain without becoming owned by LIGA™ and without converting that external governance object into Liability.

Canonical Boundary

LCXCM begins with a:

Governed Liability Connection Relationship Set

and asks:

How do these Relationships combine into complete, ordered, traceable, and materially coherent Liability Connection Chains?

LCXCM governs:

Chain Formation, Chain Identity, Chain Origin, Chain Endpoint, Direction, Sequence, Intermediate Elements, Intermediate Relationships, Direct Chains, Indirect Chains, Multi-Step Chains, Parallel Chains, Converging Chains, Diverging Chains, Branching Chains, Merged Chains, Nested Chains, Dependent Chains, Alternative Chains, Competing Chains, Partial Chains, Broken Chains, Unknown Segments, Inferred Segments, Circular Structures, Connection Networks, Temporal Structure, Chain Boundaries, Chain Integrity, Chain Completeness, Change, Correction, Reassessment, Versioning, Splitting, Merging, Historical Reconstruction, Re-Entry, and Chain Sufficiency.

LCXCM does not determine:

Connection Qualification, Connection Evidence Sufficiency, Operational Causation itself, Harm itself, Accountability itself, Liability Attachment, Liability Determination, Liability Scope, Liability Allocation, Liability State, Liability Transfer, Recourse, Resolution, Discharge, or Finality.

LCXCM ends when a:

Governed Liability Connection Chain Set

is sufficiently structured for Liability Connection Qualification governance.

Use Case 1 — Autonomous Industrial System Chain

An autonomous industrial system performs an action involving a controlled machine.

LCXRM has established individual Relationships including:

Operator → Autonomous System Autonomous System → Execution Decision Execution Decision → Machine Action
Machine Action → Operational Event Operational Event → Equipment State LCXCM determines whether and how these Relationships combine into a governed Connection Chain:

Operator → Autonomous System → Execution Decision → Machine Action → Operational Event → Equipment

State

If a materially relevant Human Override state exists between the system decision and execution, LCXCM preserves it as an Intermediate Element rather than compressing the Chain.

The completed Chain still does not establish Liability.

It becomes an input to Connection Qualification.

Use Case 2 — Multi-Party Maintenance Connection Network

A technical installation involves an owner, operator, maintenance provider, and component producer.

LCXRM has established multiple relevant Relationships.

LCXCM determines that the matter cannot be represented reliably by one linear chain.

Instead, it may contain:

Maintenance Provider → Maintenance Action → Equipment State → Operational Event and:

Operator → Operational Control → Equipment Operation → Operational Event and:

Producer → Component → Equipment State These Chains converge on related operational states while retaining separate origins and intermediate Relationships.

LCXCM preserves them as a Liability Connection Network rather than collapsing them into one assumed chain.

LCXQM subsequently determines which Chains are sufficiently qualified for downstream Liability governance.

Canonical Closing Statement

Individual Relationships do not explain the complete connective structure of a Liability matter.

They must be assembled without hiding intermediate elements, reversing direction, collapsing distinct paths, or filling missing links with assumptions.

A valid Connection Chain preserves how the relevant elements actually relate across the governed structure while allowing uncertainty, alternative paths, and incomplete segments to remain visible.

A Relationship defines one connection. A Chain shows how the connections continue.

Construct the Chain before qualifying the Connection.

Governed Links

Liability Connection Standard (LCNS)

LIGA™ Architecture Map

LCXQM — Liability Connection Qualification Module

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